

Ayesha Firdaus

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PROFESSIONAL SUMMARY

Analyst with a biopharma operations background and a self-built data skillset across SQL, Python, Power BI, and ML. Built four end to end projects spanning fraud detection, dynamic pricing, demand forecasting, and inventory optimization each with measurable business outcomes

SKILLS

Languages & Querying: SQL, Python (Pandas, NumPy, Scikit-learn, Matplotlib)

BI & Visualization: Power BI, Excel, Looker Studio, Metabase

ML & Modeling: LightGBM, XGBoost, Regression, Classification, Time Series Forecasting, Hypothesis Testing, Feature Engineering

Data & Tools: Data Cleaning, Data Modeling, EDA, ETL/ELT, Git, Google BigQuery

WORK EXPERIENCE

MSAT Intern

09/2024 – 05/2025 | Bengaluru, India

Syngene International

- Monitored and analysed batch manufacturing data across multiple client projects tracked parameter distributions, flagged outliers, and reported findings to manufacturing and stakeholder teams; corrective actions led to batch purity improving from ~90% to 98% on one key protein project
- Built Excel based analysis workflows for parameter tracking across batches plotted distributions, identified process deviations, and supported root cause investigations for out of range values
- Prepared base documents and TTDs covering raw materials, consumables, quantities, vendor conditions, and specifications; coordinated with QC teams to validate inputs and close documentation loops
- Supported scale up calculations for buffer preparations and raw material quantities during lab-to-manufacturing transitions across multiple active projects
- Worked across projects for clients including BMS, Zoetis, and others in a fast moving MSAT environment, handled documentation, cross-functional coordination, and ad hoc analytical requests in parallel

PROJECTS

Banking Fraud Detection & Transaction Risk Analytics 🔗

03/2026 – 04/2026

SQL Server, Python, XGBoost, Power BI

- Designed a four-layer fraud detection system: SQL analytics, behavioral risk scoring, XGBoost (ROC-AUC 0.844), and Power BI dashboards across 44K+ transactions; identified \$380M in fraud exposure within the dataset
- Custom risk engine stratified transactions into four bands; Critical Risk segment isolated at 60%+ fraud rate, flagging 665 high-priority customers for investigation
- Key fraud signals: crypto merchants (40.1% fraud rate), 1AM peak transactions (23.4% fraud rate), and fraud occurring ~6× farther from customer home locations surfaced via 4-page executive dashboard

Customer Channel Affinity Prediction System 🔗

04/2026 – 05/2026

Python, LightGBM, Streamlit

- Predicted each customer's most likely marketing channel using 140+ behavioral features across 9 data sources
- Implemented leakage-safe temporal validation; LightGBM classifier achieved 33.5% top-1 / 70.4% top-3 accuracy (2.3× lift over random vs 14.3% baseline)
- Deployed full prediction pipeline as a live Streamlit app with customer level channel recommendations, probability rankings, and budget allocation simulator

Dynamic Pricing & Demand Forecasting System 🔗

03/2026 – 05/2026

Python, LightGBM, Streamlit

- Built an end to end ML pipeline forecasting hourly demand and computing revenue-optimal dynamic prices
- Engineered 23 features on 8,737 hourly records using lag windows, DSR, and event flags
- Achieved 3.1% MAPE vs 16.7% naive baseline, delivering +12.7% revenue uplift (₹1.45M annualised) with 83.7% demand retention

Supply Chain Inventory Optimization & Demand Forecasting 🔗

04/2026 – 05/2026

SQL Server, Python, Power BI

- Built an end-to-end analytics pipeline diagnosing inventory inefficiencies across 6 supply chain tables, engineered reorder points, safety stock, and turnover KPIs in SQL; built 30-day rolling demand forecasts in Python
- Identified that ~95% of products were overstocked, with total inventory held at approximately 23× forecasted monthly demand, surfaced via a 3-page executive Power BI dashboard with product level drill down and supplier delay analysis
- Flagged Electronics as the lowest turnover, highest-priority category for inventory reduction; quantified a potential reduction of ~196 units per product on average against current avg stock of ~221

EDUCATION

Vellore Institute of Technology

09/2020 – 08/2025 | Vellore, India

MSc. Integrated Biotechnology

CGPA - 8.31/10

Relevant coursework: Statistics, Data Analysis, Research Methodology

AnalytixLabs & E&ICT Academy, IIT Guwahati

08/2025 – Present | Bengaluru

Advanced Certification in Data Science (ACDS)